

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2021
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $2 + 2 \times 2 - 1$. [2]
2. Evaluate $-12 - (-14)$. [3]
3. Work out $1 + \frac{2}{3}$. Give your answer as a mixed number if possible. [3]
4. Find 15% of 200. [3]
5. Solve $6x + 2 = 56$. [3]
6. Share \$22 in the ratio 4 : 7. [4]
7. Find (a) the HCF and (b) the LCM of 18 and 20. [4]
8. Write 50000 in standard form. [3]
9. Simplify $2^4 \times 2^4$. Leave your answer as a power of 2. [3]
10. $n(A) = 10$, $n(B) = 14$ and $n(A \cap B) = 4$. Find $n(A \cup B)$. [3]
11. Expand and simplify $3(x + 3) - 2x$. [3]
12. Factorise completely $18x + 9$. [3]
13. Factorise $x^2 + 8x + 16$. [4]
14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. [4]
15. Solve $4x + 3 < 19$. [3]
16. Solve the simultaneous equations
$$\begin{aligned}x + y &= 8 \\x - y &= -2.\end{aligned}$$
[4]
17. The n th term of a sequence is $5 + 2(n - 1)$. Find the 8th term. [3]
18. In triangle ABC, angle A = 58° and angle B = 52° . Find angle C. [3]
19. A regular polygon has 6 sides. Find the size of each exterior angle. [3]
20. A right-angled triangle has shorter sides 9 cm and 12 cm. Calculate the hypotenuse. [4]
21. In right-angled triangle PQR, angle P = 30° and hypotenuse PR = 12 cm. Calculate PQ, the side opposite 30° . [4]

- 22.** A triangle has base 10 cm and perpendicular height 7 cm. Find its area. **[3]**
- 23.** Calculate the area of a circle of radius 7 cm. Take $\pi = 22/7$. **[4]**
- 24.** A cylinder has radius 7 cm and height 6 cm. Find its volume. Take $\pi = 22/7$. **[4]**
- 25.** Find the gradient of the line joining (5, 2) and (7, 8). **[3]**
- 26.** Find the midpoint of the line joining (5, 2) and (7, 8). **[3]**
- 27.** A bag contains 3 red, 4 blue and 5 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**
- 28.** Find the mean of 11, 7, 4, 6, 11. **[3]**
- 29.** Work out $\begin{pmatrix} 1 & 2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix}$ where $\begin{pmatrix} \\ \end{pmatrix}$ denotes a 2×2 matrix times a column vector. **[4]**
- 30.** A trader in Mbare buys a crate of tomatoes for \$80 and sells it for \$88. Calculate the percentage profit. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).